
UNIT 1 INTRODUCTION TO MOBILE COMMUNICATIONS

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1.0 INTRODUCTION

If we have to predict the transition of computers in the next decade based on the transition in the last two decades, our imagination will probably run out.

Mobile communication has known no bounds in becoming more efficient both in terms of time and money. Earlier there was no mobility in communication. If suppose a person had to make a call, he/she was stuck by the side of a wired device to do so. Today we can make a call through a smartwatch or even a voice command. By this premise, it is unimaginable how flexible communication will get in the coming decade.

In this Unit1, we will look upon the different advances in communication over the years. Even though a lot of techniques get outdated, their basis are often used to make enhanced devices for communication.

A communication device often shows one of the following features:

Fixed and wired: For example a desktop PC. It is not applicable for mobile usage due to its weight and power consumption.

Mobile and wired: For example, a laptop can be carried from one place to another and can connect to any network.

Fixed and wireless: For example, installing networks, like those in historical buildings. It is done to avoid any damage that might be caused by installing wires.

Mobile and wireless: For example, GSM, which has no cable restrictions and can travel between different wireless networks.

1.1 OBJECTIVES

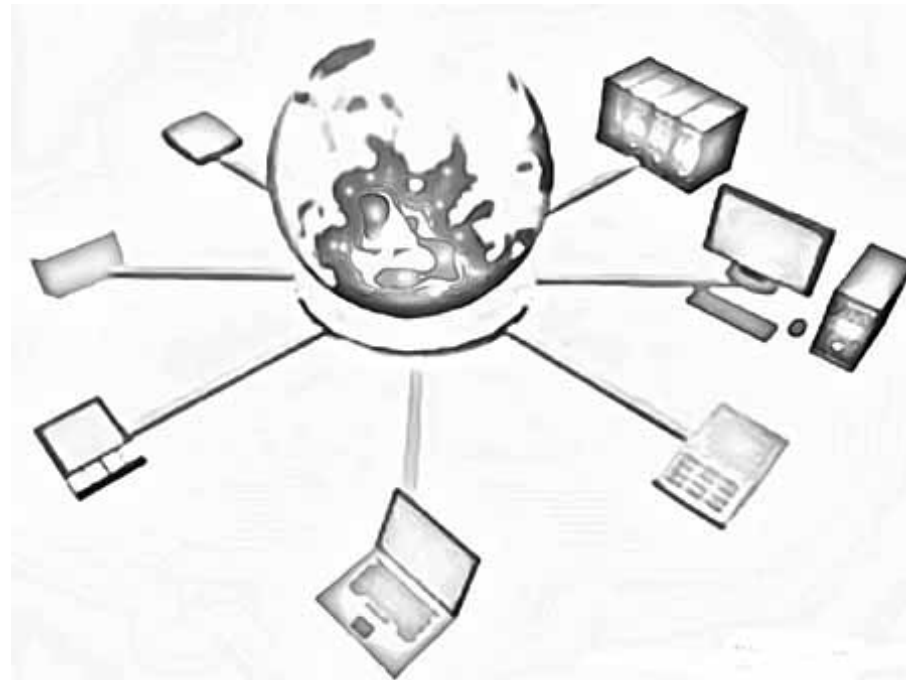
At the end of this Unit, you shall be able to:

- (i) understand the concepts of Mobile Computing, how it works and the various features of Mobile Computing.

- (ii) Understand multiplexing concepts, its different type, its advantages, disadvantages and applications.
- (iii) Understand how GSM works and its advantages and disadvantages.
- (iv) Understand how GPRS works and what are its features.
- (v) Understand how 3G works and learning its advantages and disadvantages.
- (vi) Understand how 4G works and learning its advantages and disadvantages.
- (vii) Understand the difference between LTE and WIMAX.

1.2 MOBILE COMPUTING

Mobile Communication refers to the infrastructure put in place to facilitate and support the stated services with regard to seamless and reliable communication. Devices such as protocols, services, bandwidth, and portals are part of this infrastructure. At this stage, the format of the data is defined. By doing so, a collision with another system offering the same service is avoided.



Essentially, the overlaying infrastructure is radio wave-oriented since the media is unguided/unbounded. Therefore, the signals are transmitted over the air to devices that can receive and send the same types of signals.

1.2.1 User Mobility

An individual who has access to the same or similar telecommunications services at different locations is referred to as a mobile user. The user can move between different geographical

locations, networks, communication devices and different applications.

1.2.2 Device Mobility

A number of mechanisms are present in both the device and the network that ensure that communication is still possible even while the device is moving. The device moves between different geographical locations and networks.

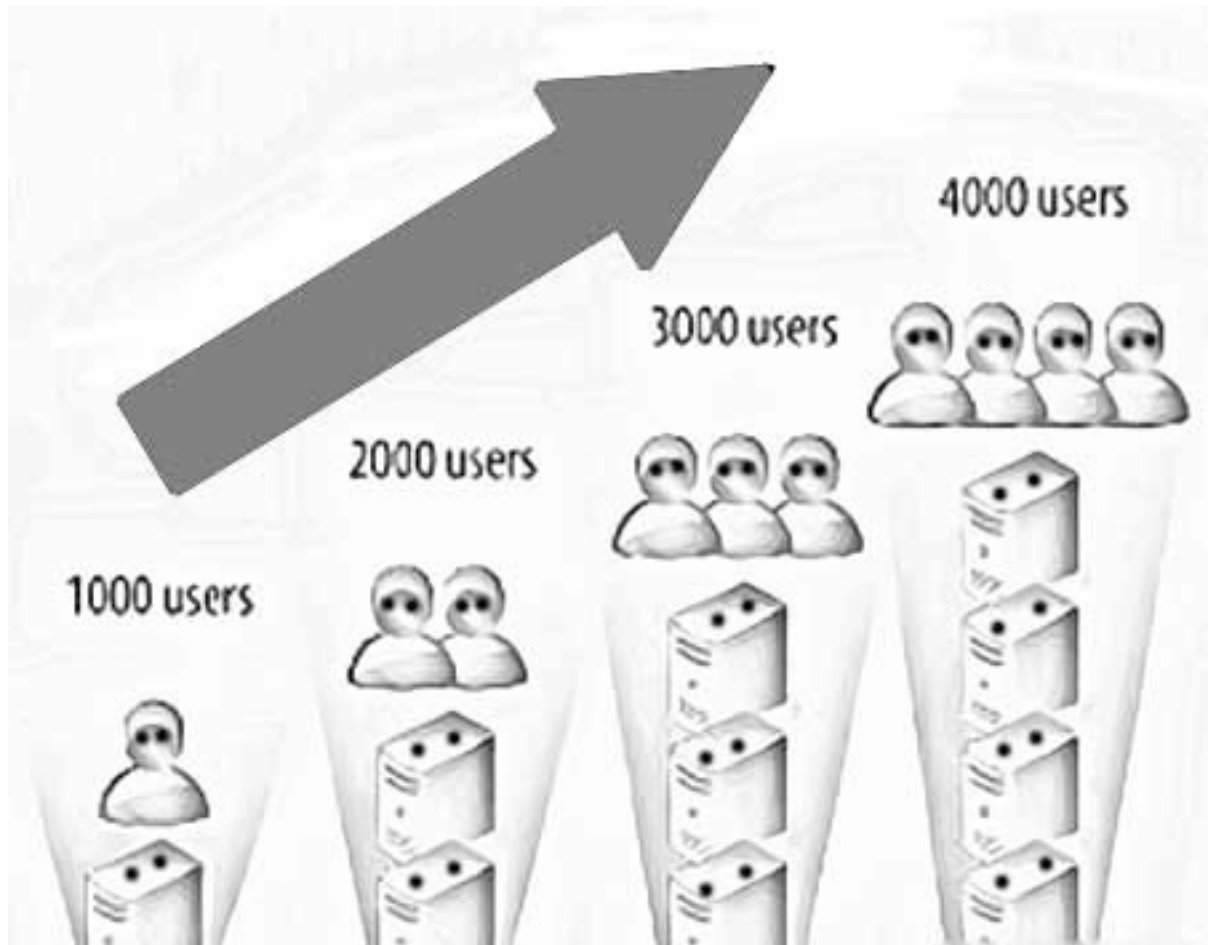
1.3 FEATURES OF MOBILE COMMUNICATION

A high capacity load balancer is essential for all wired and wireless infrastructures.

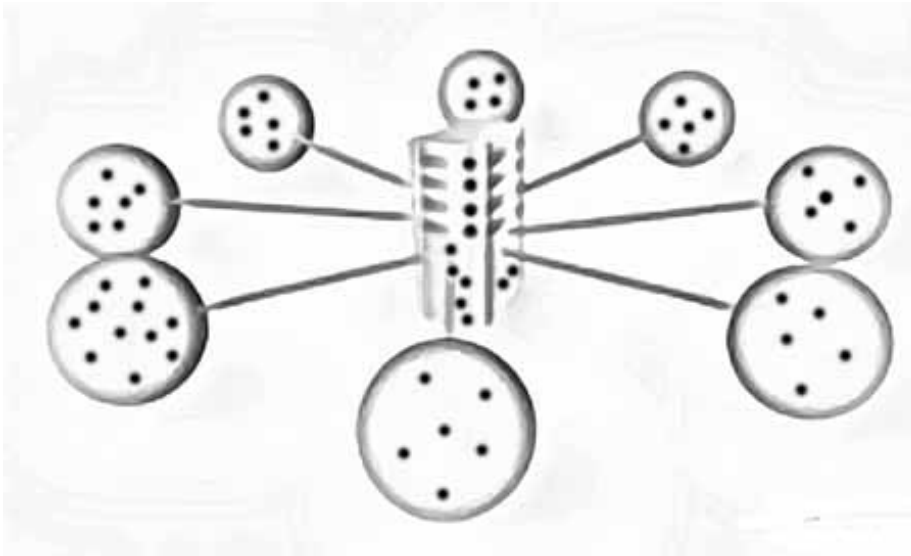
When one access point is overloaded, the system will automatically choose a different access point based on the available capacity.



1.3.1 Scalability: Wireless devices continue to grow in popularity every day. As long as coverage and capacity are not overextended, the wireless networks can expand as requirements change - without having to rework or re-create the entire network.



1.3.2 Network management system: There are now a great many different components in wireless networks, such as access points, firewalls, switches, and managed power. Wireless networks offer a better way to manage the entire network.



1.3.3 Role based access control: Using role-based access control (RBAC), you are able to assign roles based on what, who, where, when and how a user or device is trying to access your network. Access control rules or policies can then be enforced based on the ends or roles of the device.

1.3.4 Provide outdoor and indoor coverage options: Your wireless system should provide outdoor and indoor coverage. It is important that a secure registration process be used for network access control, also known as mobile device registration.

In addition to controlling the role of each user and enforcing policies, network access control can be configured to allow users to register themselves with your network.

1.3.5 Manage mobile devices: Imagine a scenario in which thousands of mobile devices access your wireless network and run thousands of applications.



1.3.5.1 Roaming: As you move throughout your office or even from one building to another, you won't experience dropped connections, slower speeds or any interruptions in service.

1.3.5.2 Redundancy: Depending on your specific needs and environment, your wireless system may require varying levels of redundancy.

In order to maintain proper security, you must use the right firewall. Your network firewall is the system's backbone. Having the right firewall in place will help you in many ways:

- (a) Apps and users can be viewed and controlled.
- (b) Reduce complexity by Antivirus protection. Deep Packet Inspection (DPI) Filtering applications.
- (c) Ensure your network and users are protected against known and unknown threads including: zero-day, encrypted malware, ransomware, malicious botnets.

1.4 WHAT IS MULTIPLEXING?

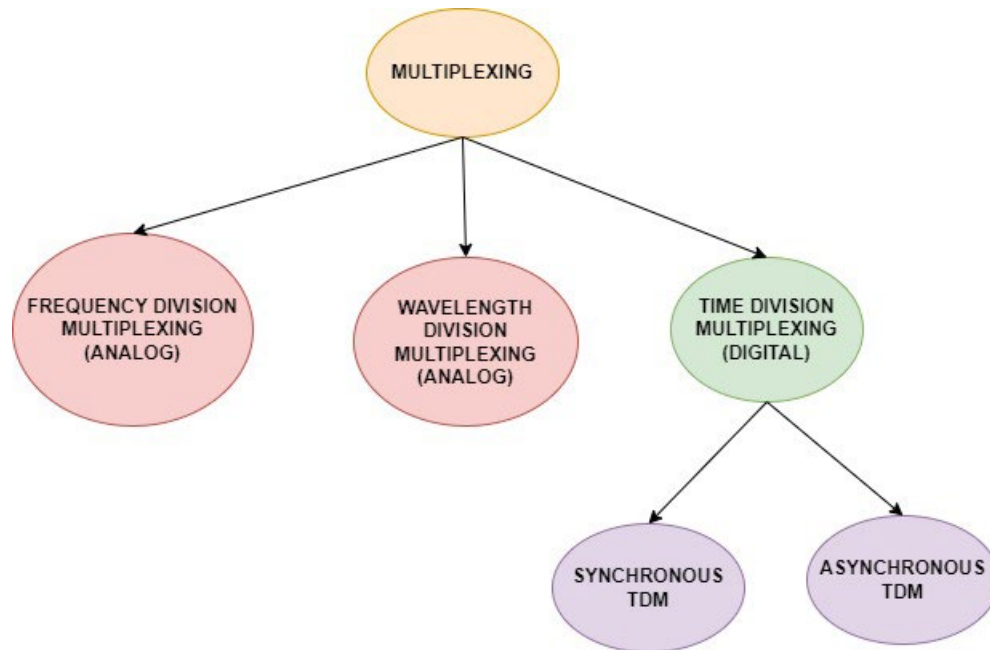
As a technique in electronics and signal processing, multiplexing is a method of combining multiple analog or digital signals into one over a shared medium. In mobile computing, communications, and computer networks, multiplexing can be used to combine analog or digital signals.

The means of communication can include radio frequency (radio) and cable (cable). All media can be multiplexed.

An example of multiplexing is how multiple calls can be connected through one telephone wire.

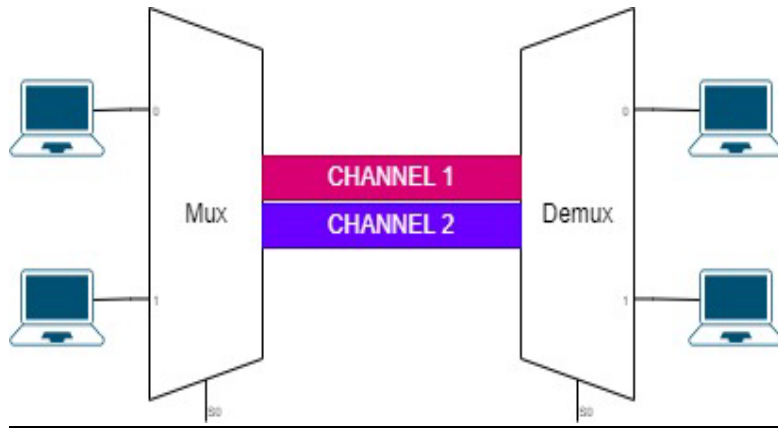
Types of Multiplexing

1. Frequency Division Multiplexing (FDM)
2. Time Division Multiplexing(TDM)
3. Code Division Multiplexing(CDM)



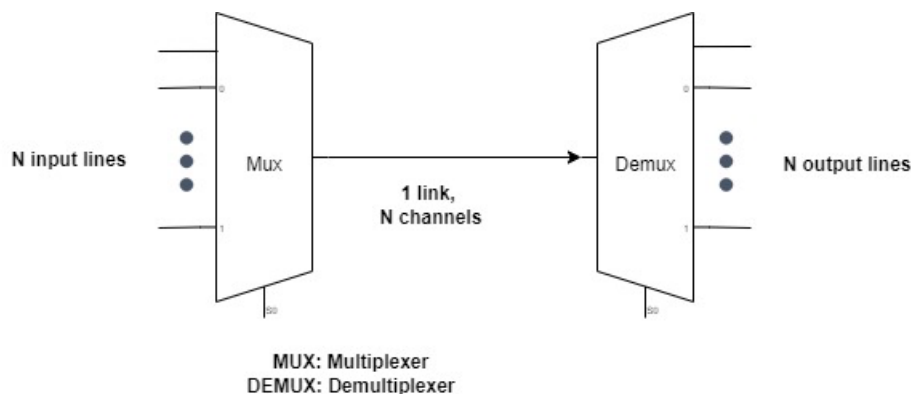
1.4.1 Frequency Division Multiplexing (FDM)

There are several signals being transmitted simultaneously, and each source transmits its signals within the allotted frequency range. The two adjacent signals are separated by a suitable frequency gap to avoid overlapping. Due to the signals being transmitted at the allotted frequencies, the chances of collisions are decreased. Several logical channels exist in the frequency spectrum, each of which corresponds to a particular bandwidth. Several signals are sent simultaneously at the same time, each with a different frequency band or channel. Such transmissions are used in radio and television. Guard bands are therefore used to avoid interference between successive channels.



1.4.1.1 Advantages of FDM:

1. Easy to implement.
2. Efficient even when traffic is constant.
3. No equalization required.
4. It can be possible to increase capacity by reducing the knowledge bit rate and using efficient digital codes.
5. It is not difficult to implement technological advances. For instance, systems are often designed so that improvement in terms of speech coders can be incorporated easily.
6. Since FDMA systems use low bit rates (large symbol time) in comparison to average delay spread, they are less expensive and exhibit low Inter Symbol Interference (ISI).



1.4.1.2 Disadvantages of FDM:

1. Guard Bands, although necessary, can also be seen as a wastage of capacity.
2. Network planning is time critical.
3. Low capacity of traffic.
4. Requires expensive multi-channel receivers.
5. The system may be more expensive if RF (Radio Frequency) filters must meet stringent specifications for adjacent channel rejection.

6. Due to the fact that channels are assigned for a single user, there are idle channels in a general system.

1.4.1.3 Applications of FDM:

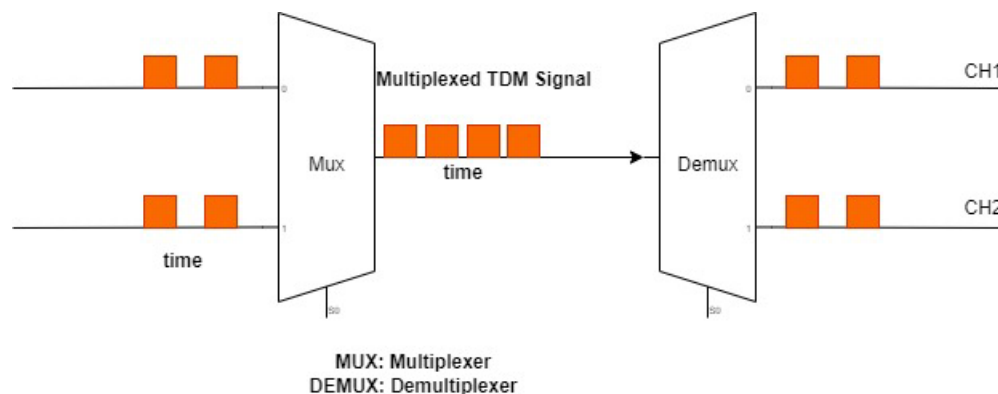
FDM is used in:

1. Telephone system.
2. FM & AM Radio broadcasting.
3. Cable TV.
4. Walkie talkies.
5. Mobile network for a closed user group. e.g. Wi-Fi.
6. Total access communication systems (TACS)
7. 2G mobile communication.

1.4.2 Time Division Multiplexing(TDM)

Time Division Multiplexing (TDM) is a multiplexing technique that enables multiple data signals to be carried in different time slots over a common communication channel.

One frame is said to be transmitted when its entire signal gets transmitted across the channel. Time slots are used to divide the overall time domain into multiple fixed length time slots.



1.4.2.1 Synchronous TDM

In Synchronous TDM, each time slot is pre-assigned to a constant source.

It is based on the assumption that data will be present at the source regardless of how the slots are allocated in advance. This results in wastage of channel capacity since time slots are completely wasted in the absence of data.

1.4.2.2 Asynchronous TDM

Asynchronous TDM, also known as statistical division multiplexing, is a method in which time slots are allocated only to machines that have the required information to send data.

1.4.2.3 Advantages of TDM:

1. TDM is very flexible.
2. The circuitry is not as complex,
3. Cross talk problem is not so severe.
4. The channel bandwidth can be completely utilised for each channel.

1.4.2.4 Disadvantages of TDM:

1. Synchronization is required.
2. The implementation is complex.
3. Slow narrowband fading may wipe out all TDM channels.

1.4.2.5 Applications of TDM:

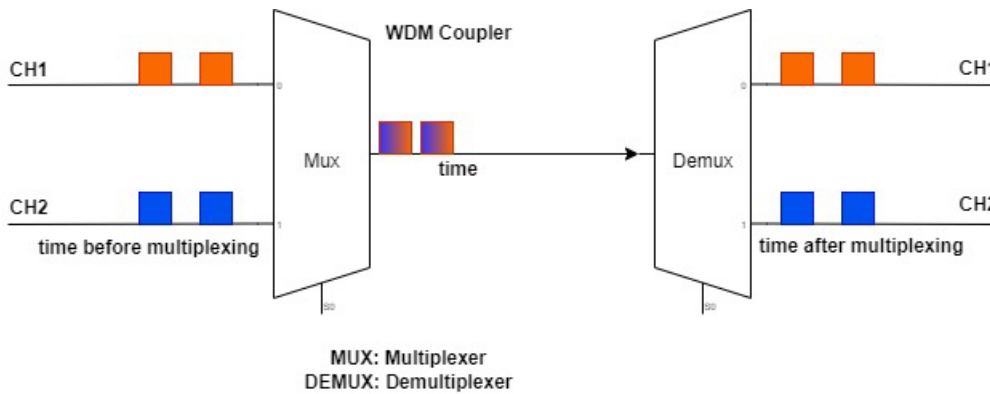
1. It is used in SONET (Synchronous Optical Networking).
2. It is deployed in a public switched telephone network.
3. It's found in telephone wired lines.
4. It is used in digital audio mixing systems.
5. It is used in a half duplex communication system.
6. It is used in GSM.

1.4.3 Wavelength Division Multiplexing

Wavelength Division Multiplexing (WDM) is a networking technique that allows multiple data signals to be simultaneously transmitted over a common frequency band.

The technology of WDM is used when multiple users are permitted to share one communication channel.

As multiple users transmit within the same frequency spectrum, WDM provides some level of security. To decode each transmission, the spreading code must be used.



1.4.3.1 Advantages of WDM:

1. The WDM channel is not effectively decodable, thus it provides enhanced protection for cell communication.
2. Due to the greater number of clients per MHz of data transmission, it has multiple times the limit of GSM so it provides better inclusion and requires fewer reception devices. It also burns-through less force to expand client limits.
3. The voice quality is just as good as the sign quality.
4. There is flexibility in the asset designation.
5. It is incredibly efficient.
6. There is no need for synchronization with WDM.
7. It can transmit data to a large number of clients simultaneously.

1.4.3.2 Disadvantages of WDM:

1. It requires time synchronization.
2. As the number of clients increases, WDM framework execution degrades.
3. As a WDM user, choosing the code length can be a laborious process, since it can cause delays.
4. The quality of administration diminishes when the amount of clients expands.
5. When we use WDM strategies, we encounter the close-far issue, Self-sticking issue.

1.4.3.3 Applications of WDM:

1. It is used for military and commercial applications.
2. It is implemented in mobile communications.
3. It is also used in radar and navigation systems.

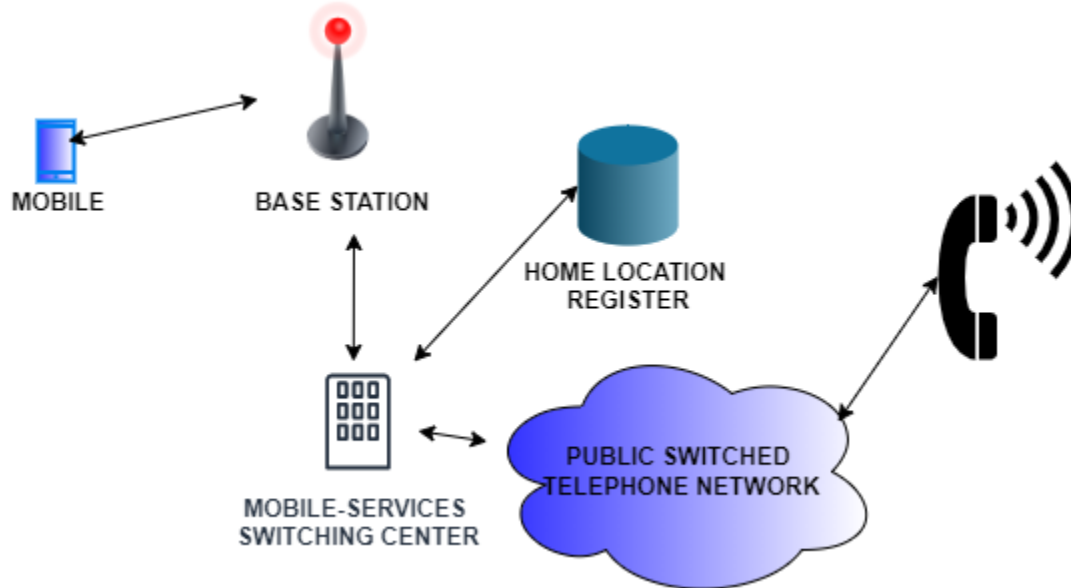
1.5 GSM(GLOBAL SYSTEM FOR MOBILE COMMUNICATION)

A mobile phone uses GSM technology, which is an open and digital cellular protocol. There are four different frequency bands used by this technology, including 850 MHz, 900 MHz, 1800 MHz, and 1900 MHz. It combines FDMA and TDMA.

During the early 1970s, Bell Laboratories developed a cell-based mobile radio system that formed the basis for GSM.

Four separate components make up GSM networks:

- The mobile phone
- The base station system (BSS)
- The network switching system (NSS)
- The operation and support system (OSS)



The base station system (BSS) carries data between the cellphones and the network system servers. It comprises two main components: the base transceiver station (BTS) and the base station controller (BSC).

In GSM networks, the NSS, also known as the core network, tracks the location of callers in order to provide cellular services. It includes a mobile switching centre (MSC) and a home location register (HLR). A mobile service switching centre is like a standard ISDN or PSTN switching node

1.5.1 Advantages of GSM:

1. GSM provides worldwide roaming for its customers.
2. Due to its unique devices and facilities, GSM is extremely secure.
3. Spectrum efficiency and clear voice calls.
4. You can send short messages, view your caller's ID, place a call on hold, and forward your calls.
5. AD is available on Integrated Services Digital Network (ISDN) and other telephone company services.

1.5.2 Disadvantages of GSM:

1. As multiple users share the same bandwidth, bandwidth lag can occur.
2. Due to the interference that is caused by pulse transmission technology, some electronics, such as hearing aids, cannot be used in certain places, such as hospitals, airports, and petrol pumps.
3. Repeaters are needed to increase coverage.
4. The maximum call range of GSM is 35 km.
5. The data of users is not encrypted end-to-end.

EXERCISE 1

Question 1: How do you define mobility in mobile computing?

Question 2: What are the uses of mobile computing?

Question 3: List the advantages of mobile computing.

Question 4: Describe multiplexing and its uses.

Question 5: What are the different types of multiplexing?

Question 6: Describe in detail the advantages and disadvantages of the different types of multiplexing.

Question 7: What is the full form of GSM?

Question 8: List the components of GSM.

Question 9: List the advantages and disadvantages of GSM.

1.6 GPRS AND 2.5G

As part of GSM's global system for mobile communications (GSM), General Packet Radio Service (GPRS) is a packet-oriented mobile data standard. As a response to CDPD and i-mode packet-switched cellular technology, the European Telecommunications Standards Institute (ETSI) developed GPRS. The 3GPP is now responsible for maintaining it.

A GPRS network is a packet-switching communications protocol that uses the best-effort packet switching method.

The technology allows mobile data to be transmitted and received more efficiently, more quickly, and more cheaply. Mobile devices using GPRS are always connected to the internet, making them always online and liable for data usage charges. When used, the device only keeps the connection busy. So, capacity is better utilized, and more data can be exchanged simultaneously. There is a maximum speed range of 7 KB/s to 14 KB/S on GPRS. It is called **2.5G**, which is the state of wireless technology between the second and third generations of wireless technology: before and after General Packet Radio Services (**GPRS**).

A generation is not official. It is called so because it is in the middle of 2G and 3G. There is still support for 2G and 3G within M2M.

In contrast to the circuit-based switching protocols of 2G, GPRS uses packet-switching communications. Due to this, data delivery is best-effort; there can be a variation in latency and deliverability. The Quality of Service (QoS) of GPRS is difficult to manage due to the number of concurrent users.

In the past, GPRS (2.5G) speeds have been quoted over 2G networks; GPRS can theoretically transmit around 120 kilobits per second over 2G networks. In real-world conditions, you can expect speeds of 20-50 kbps. Latency may vary, but typically will range from .5 to 1 second.

1.6.1 Features of GPRS

1. It is a packet-based data network that is well suited for non-real-time Internet applications, such as retrieving email, fax messages, and asymmetric web browsing, where users download more data than they upload.
2. Multi-user GPRS networks share individual radio channels and time slots among users.
3. In contrast to HSCSD, GPRS can support many more users, but in a burst manner.
4. An IS-136 or GSM-based packet network is provided by the GPRS standard. For better packet data access, GPRS retains the original modulation formats of 2G TDMA while using a completely redesigned air interface.
5. A user can achieve as much as 171.2 kbps using all 8 Time-slots of an individual GSM radio channel.
6. Data payloads carried in GPRS must be error corrected by the applications.

1.6.2 Services offered by GPRS

By extending the GSM Packet circuit switched data capabilities, GPRS can deliver the following services:

1. SMS messaging and broadcasting
2. Multimedia messaging service (MMS)
3. Push-to-talk over cellular (PoC)
4. Instant messaging and presence—wireless village
5. Point-to-point (P2P) service
6. Through wireless application protocol (WAP), smart devices can access the Internet
7. Point-to-multipoint (P2M) service
8. If SMS over GPRS is used, an SMS transmission speed of about 30 SMS messages per minute may be achieved. This is much faster than using the ordinary SMS over GSM, whose SMS transmission speed is about 6 to 10 SMS messages per minute.

1.7 THIRD GENERATION (3G)

In the early 1980s, the International Telecommunication Union (ITU) carried out research and development on 3G technology.

This new generation of wireless technology allows handheld devices to access high-speed data and voice services. 3G networks will provide multimedia services combining voice and data.

1.7.1 3G WiFi

A device that includes cellular data service of the third generation (3G) as well as Wi-Fi is a 3G Wifi. In recent years, Apple's iPad and Amazon's Kindle readers have been among the first to offer 3G and Wi-Fi connectivity. When a mobile phone is equipped with both 3G and Wi-Fi, it is possible to access the Internet regardless of where the user is located. When the mobile phone is equipped with only Wi-Fi, the user is required to be near a Wi-Fi hotspot at home, work, or a café or hotel.

1.7.2 Advantages of 3G

1. Overcrowding in existing systems can be relieved by adding new radio spectrum.
2. The capacity, security, and reliability of the network is increased.
3. A device that is always online, 3G uses IP connectivity, which is packet-based (not circuit based).
4. As the third generation of data communication and mobile phone standards, 3G offers higher bandwidth for video and web-based applications. It offers faster data transfer and better voice quality than 2G, 2.5G, GPRS, and 2.75G EDGE networks.
5. Wireless voice, video, and data services are included in 3G Services.
6. The 3G network supports services with at least 144 kbit/s of information transfer rate. As a result of further 3G releases, known as 3.5G and 3.7G, smartphones as well as mobile modems in laptop computers can also access mobile broadband of several Mbit/s. As a result, it can be used for wireless voice calls, mobile Internet access, fixed wireless Internet access, video calls and mobile TV services.
7. There are several 3G systems, including the Universal Mobile Telecommunications System (UMTS), Wideband CDMA (W-CDMA), and Code Division Multiple Access 2000 (CDMA2000).

1.7.3 Disadvantages of 3G

1. Requires different handsets.
2. Insufficient bandwidth.
3. High power consumption.
4. It is very expensive to upgrade base stations and cellular infrastructure to 3G.
5. Closer base stations are required making it more costly.
6. Costly spectrum-licence, expensive network deployments, and subsidies for handsets.

1.7.4 3G is being phased out for what reason?

It is expected that AT&T will terminate its "third generation" telecommunications network to devote more bandwidth to developing much more advanced 5G networks. People who use 3G or some 4G devices without VOLTE (Voice over LTE) won't be able to access the Internet.

1.8 FOURTH GENERATION (4G)

In contrast to 3G mobile networks, 4G mobile phones offer broadband cellular network services. A fully IP-based cellular communication system is provided. International Telecommunication Union (ITU) specifications govern the capabilities provided by IMT-Advanced.

4G connections connect mobile devices to mobile networks via an antenna that transmits radio waves. MIMO (Multiple Input Multiple Output) and OFDM (Orthogonal Frequency Division Multiplexing) are used for 4G transmission and reception. Compared to 3G, MIMO and OFDM provide more capacity and bandwidth. TDMA (Time Division Multiple Access) and CDMA (Code Division Multiple Access) are the primary technologies that powered 3G. OFDM provides higher speed than those technologies.

1.8.1 Features of 4G Network:

1. The network provides voice, data, signals, and multimedia transmission over IP packet switches.
2. According to the IMT-Advanced specifications, high-mobility stations such as trains, cars, and residences should have peak data rates of 100Mbps, while low-mobility stations such as residences should have peak data rates of 1Gbps.
3. 4G networks must also be capable of 1 Gbps downlink over 67 MHz bandwidth or less.
4. It provides uninterrupted high quality services 24 hours a day, 7 days a week, regardless of location or time.

1.9 LONG-TERM EVOLUTION (LTE)

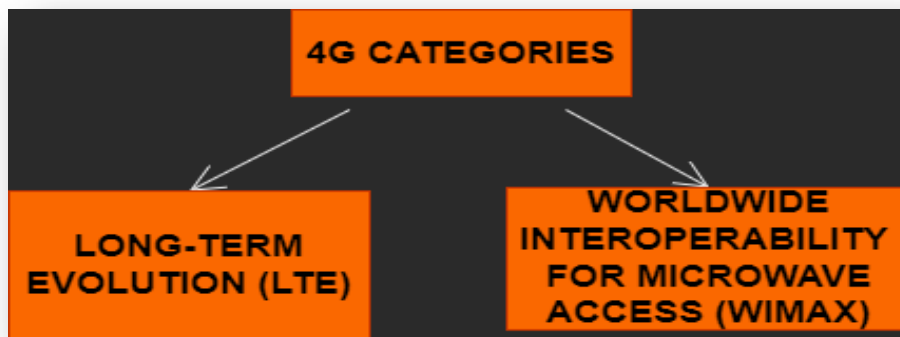
Long-term evolution (LTE) technology is an enhancement to 3G. UMTS/HSPA and GSM/EDGE technologies combine to provide high-speed mobile communication. In comparison to third-generation (3G) wireless technology, LTE (Long-Term Evolution) offers increased network capacity and speed for cellphones and other cellular devices. Compared to 3G, LTE is capable of higher peak data transfer rates, initially up to 100 Mbps downlink and 30 Mbps uplink.

LTE is often referred to as 4G LTE and is the next step in the evolution of mobile technology. It follows the specifications of 2G GSM and 3G UMTS.

According to the International Telecommunication Union (ITU), LTE is not true 4G. It was initially defined as a standard that would enable stationary users to consume a data rate of 1 Gbps and mobile users to consume 100 Mbps.

1.9.1 Features of LTE

1. In 2021, LTE's global average download speed was 17 Mbps, and its average upload speed was 12 Mbps.
2. The voice over LTE (VoLTE) technology allows users to talk without jitter or lag.
3. Compared to standard LTE, LTE-Advanced devices offer two to three times faster download and upload speeds.
4. By combining frequencies from multiple component carriers, LTE-Advanced handsets improve signal, speed, and reliability, adding bandwidth of up to 100 MHz across five component carriers (bands).



1.10 WORLDWIDE INTEROPERABILITY FOR MICROWAVE ACCESS (WiMAX)

In some cases, WiMAX is referred to as 4G because it is a standard for mobile wireless broadband access (MWBA). With 20 MHz wide channels, peak downlink data rates are 128 Mbps and uplink data rates are 56 Mbps. It is based on the IEEE 802.16 standard and is a wireless microwave technology. WiMAX forum first published it in 2001, the version that is now known as fixed WiMAX.

1.10.1 Features of WiMAX:

1. Broadband wireless access (BWA) networks will be advanced by this initiative.
2. In the initial version of WiMAX, data rates were 30-40 Mbps, but in 2011, the updated version offered 1 Gbps data rates.

3. A few similarities exist between WiMAX technology and Wi-Fi technology, hence its nickname of "Wi-Fi on steroids." However, WiMAX provides much higher data rates, is used for outdoor networks, and uses IEEE 802.16 standards rather than IEEE 802.11 standards.
4. As per user requirements, bandwidth is dynamically allocated from 2 GHz to 11 GHz.

Exercise 2

Question 1: Differentiate between GSM and GPRS.

Question 2: Explain the services offered by GPRS?

Question 3: Explain the disadvantages of 3G?

Question 4: What is the difference between 3G and 2.5G?

Question 5: Explain WiMax?

Question 6: What is the difference between 4G and 4G-LTE?

Question 7: What are the primary technologies to power 3G?

1.11 SUMMARY

A wireless-enabled computer or mobile device (or a combination of both) can transmit data, voice, and video without needing to be connected to a fixed physical network.

Mobile Computing has 2 types of mobility; user mobility and device mobility.

In mobile computing, communications, and computer networks, multiplexing can be used to combine analog or digital signals.

Multiplexing consists of three categories; TDM, FDM and WDM.

Mobile phones use GSM technology, which is an open and digital cellular protocol.

As part of GSM's global system for mobile communications (GSM), General Packet Radio Service (GPRS) is a packet-oriented mobile data standard.

3G networks provides multimedia services combining voice and data.

4G connections connect mobile devices to mobile networks via an antenna that transmits radio waves.

In comparison to third-generation (3G) wireless technology, LTE (Long-Term Evolution) offers increased network capacity and speed for cellphones and other cellular devices.

WiMAX is referred to as 4G because it is a standard for mobile wireless broadband access (MWBA).

1.12 FURTHER READINGS

1. Code Division Multiple Access Example

<https://www.youtube.com/watch?v=Ex31vvIEzZ0>

2. Mobile Computing and Wireless Communication

<https://slidetodoc.com/mobile-computing-and-wireless-communication-2170710-unit-1/>

3. Wireless Communication & Mobile Programming

<https://slidetodoc.com/wireless-communication-mobile-programming-unit-1-mobile-computing/>

4. "Mobile Computing" by Asoke K Telukder, Roopa R Yuvagal, TMH

5. FUNDAMENTALS OF MOBILE COMMUNICATION by Mehaboob Mujawar, Jafar A. Alzubi

6. Introduction to Mobile Communication

S Sureshkumar, Fr. J. Janet, APS. Anandaraj

7. Wireless And Mobile Communication by Sanjeev Kumar, New Age International (P) Ltd., Publishers

8. MOBILE AND WIRELESS COMMUNICATION

Leena R. Mehta

9. Mobile Communication

by: Behera G. K.